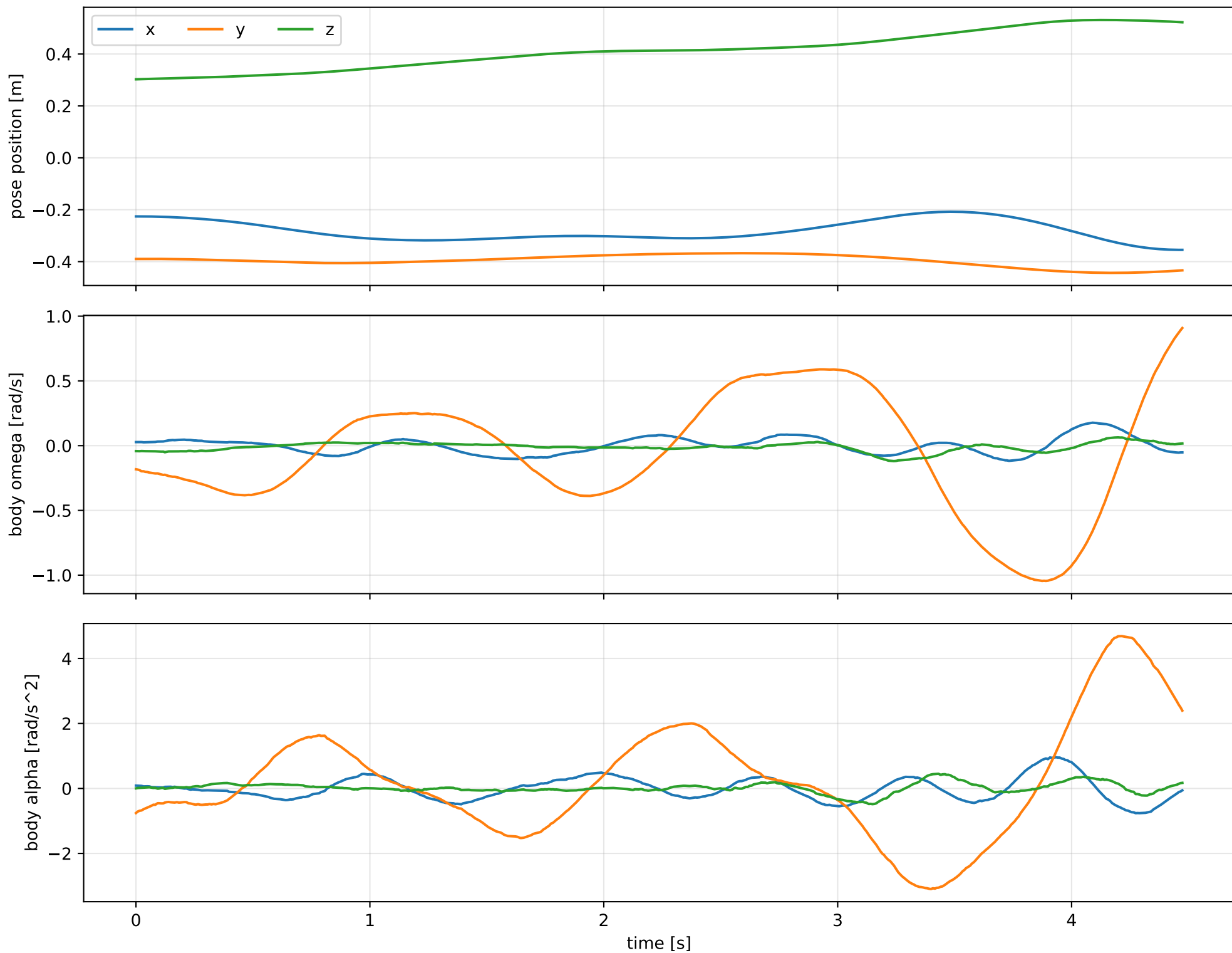
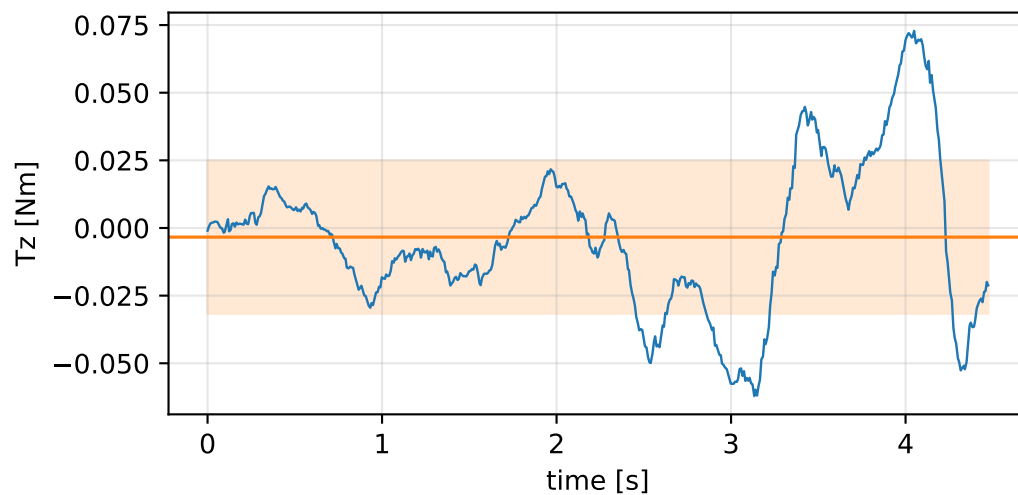
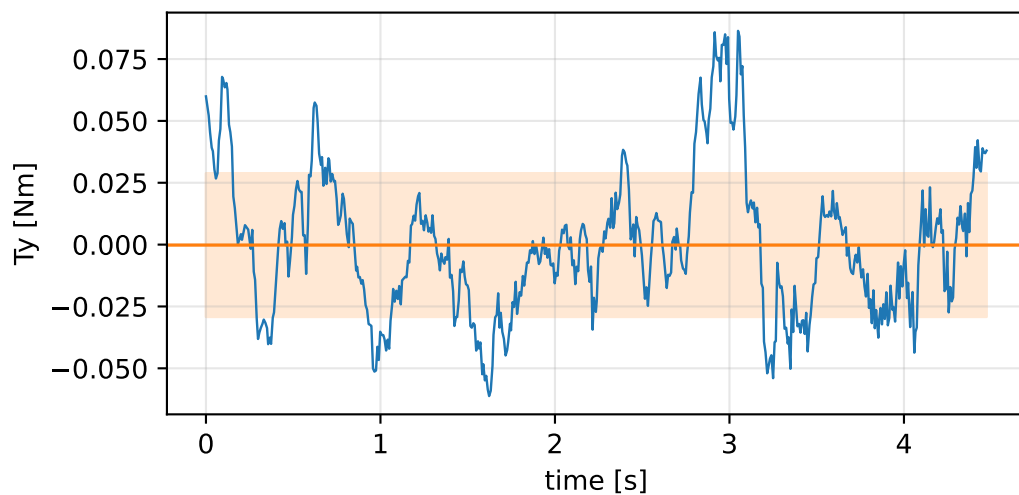
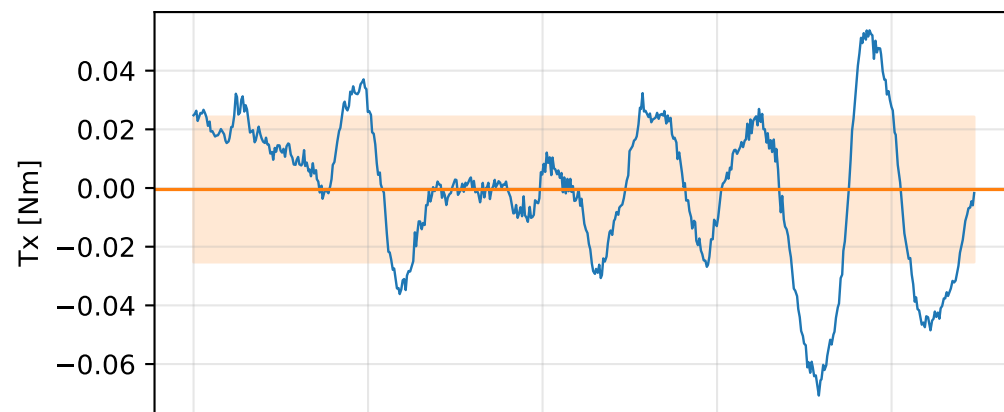
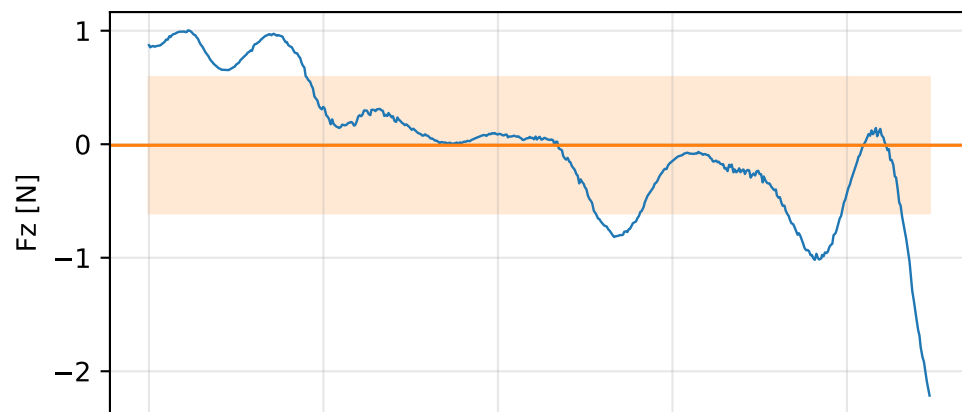
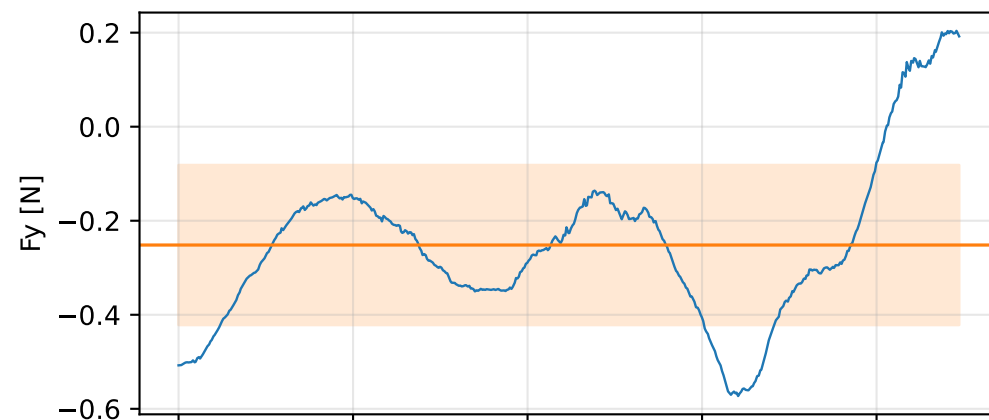
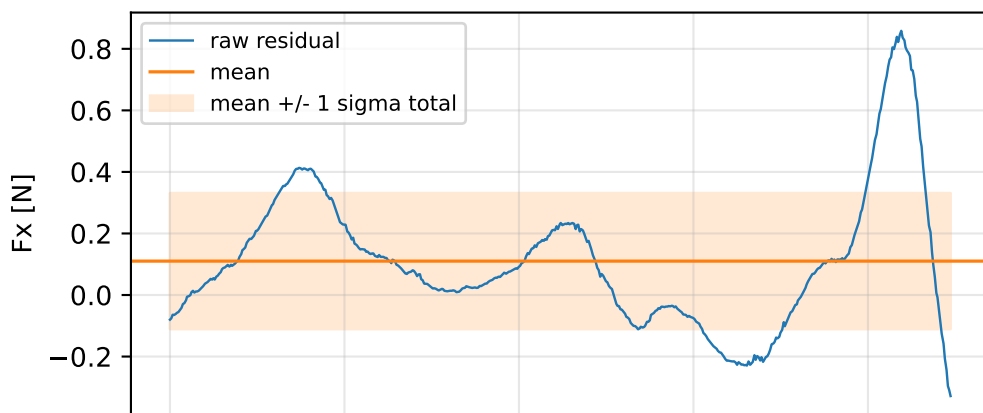


inertia_over_mass_all_nominal: SG trajectory and derived motion



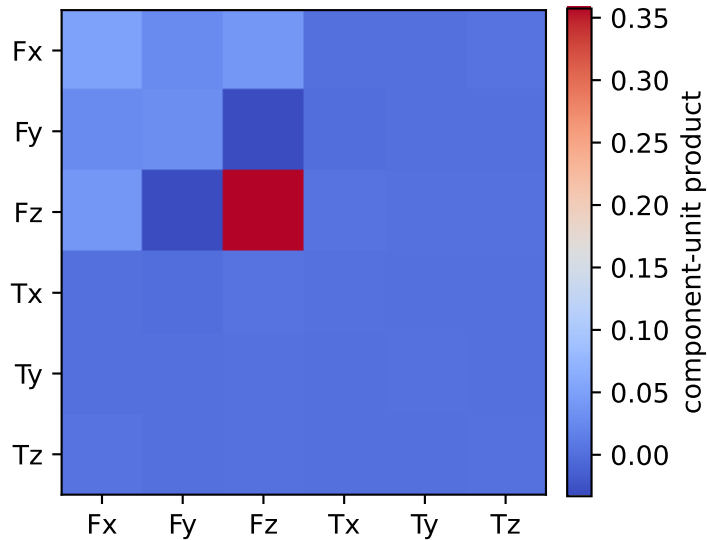
inertia_over_mass_all_nominal: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg



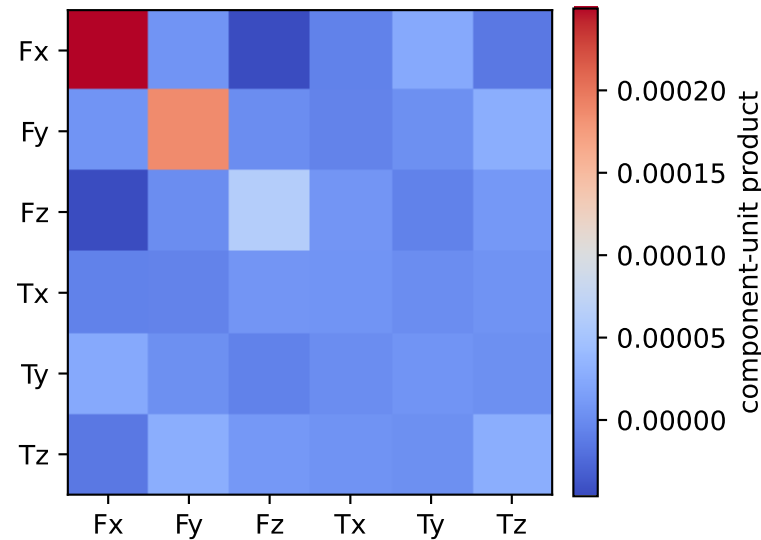
vector RMS about zero: force 0.71401 N, torque 0.047523 Nm; component std about mean: force [0.2209 0.17 0.5978], torque [0.0248 0.029 0.0282]

inertia_over_mass_all_nominal: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [0. 0.001 0.001 0.005 0.065 0.365]

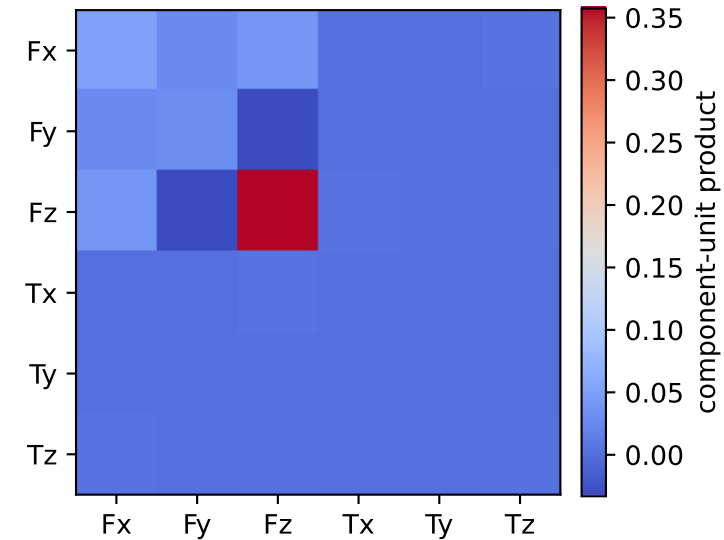
empirical total covariance



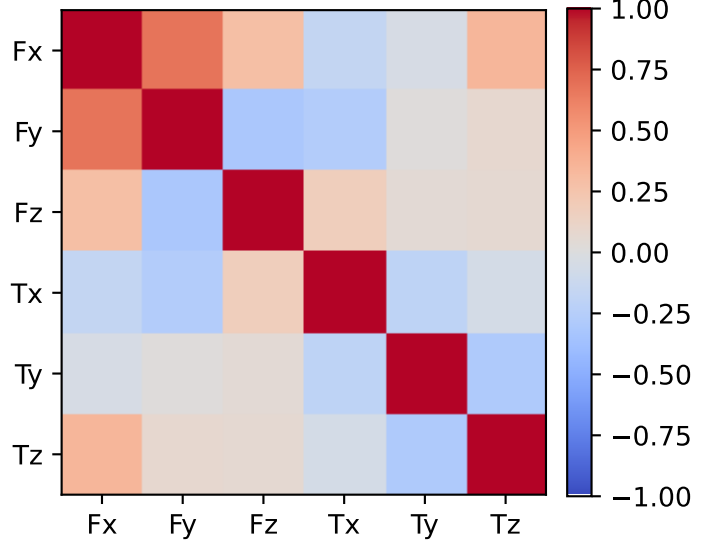
mean full-SG prediction covariance



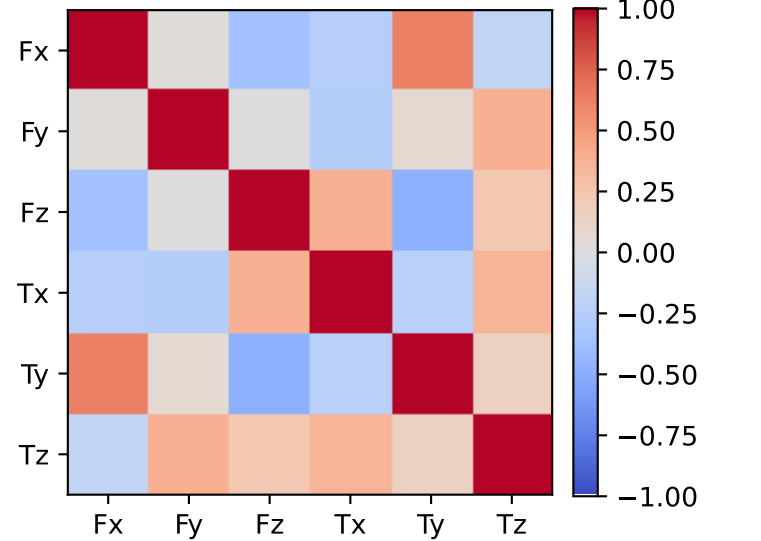
PSD excess model discrepancy covariance



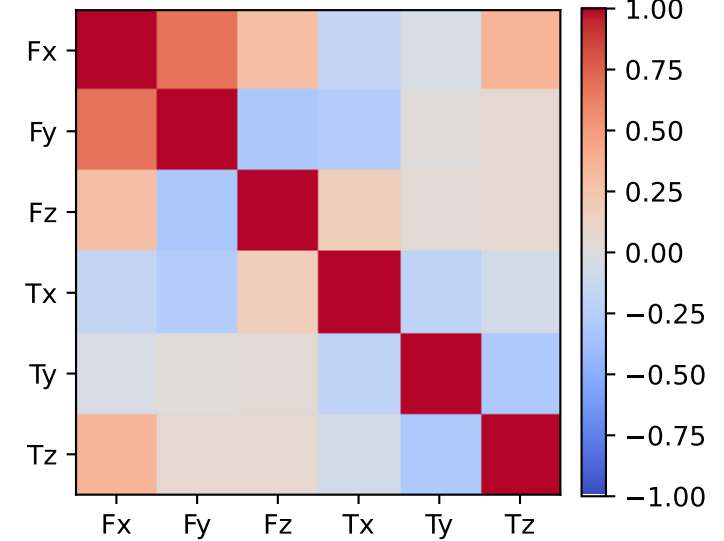
empirical total correlation



mean full-SG prediction correlation

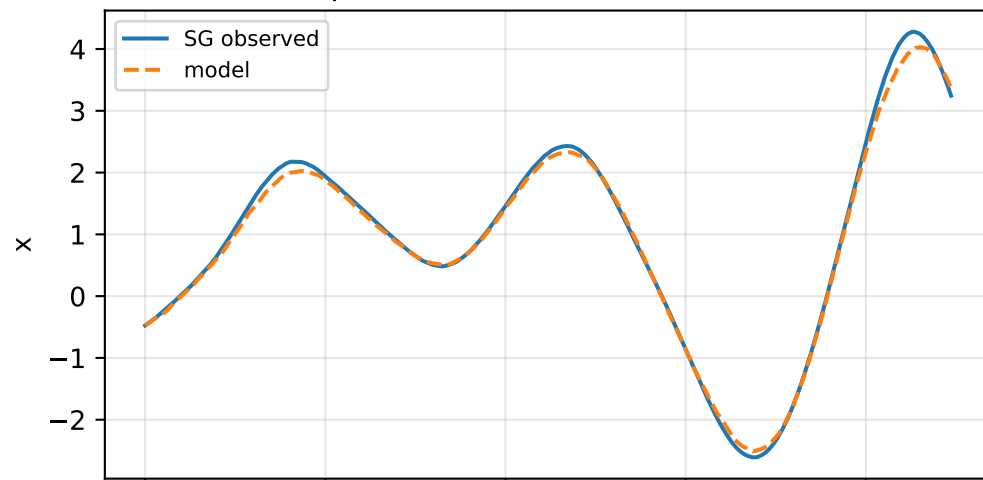


PSD excess model discrepancy correlation

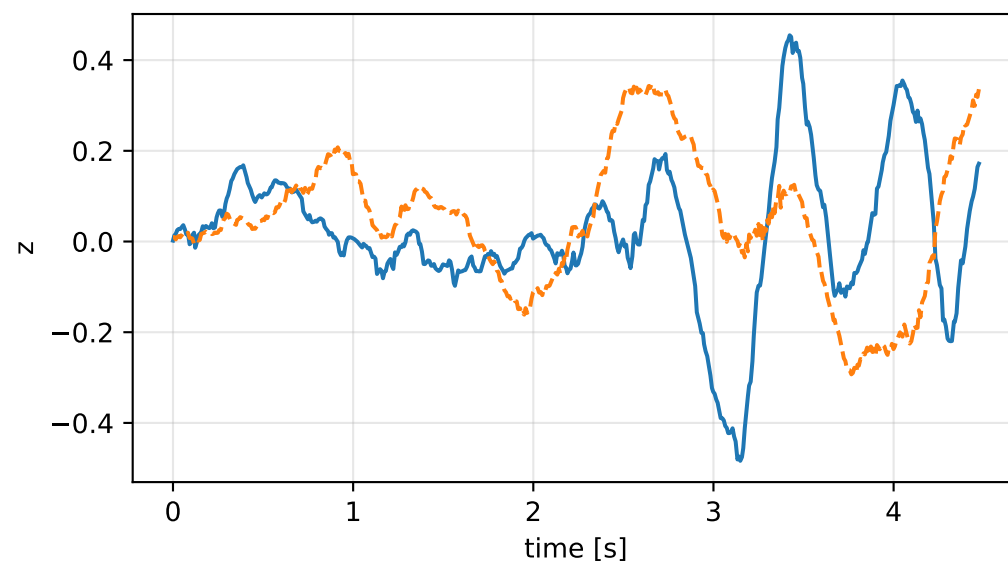
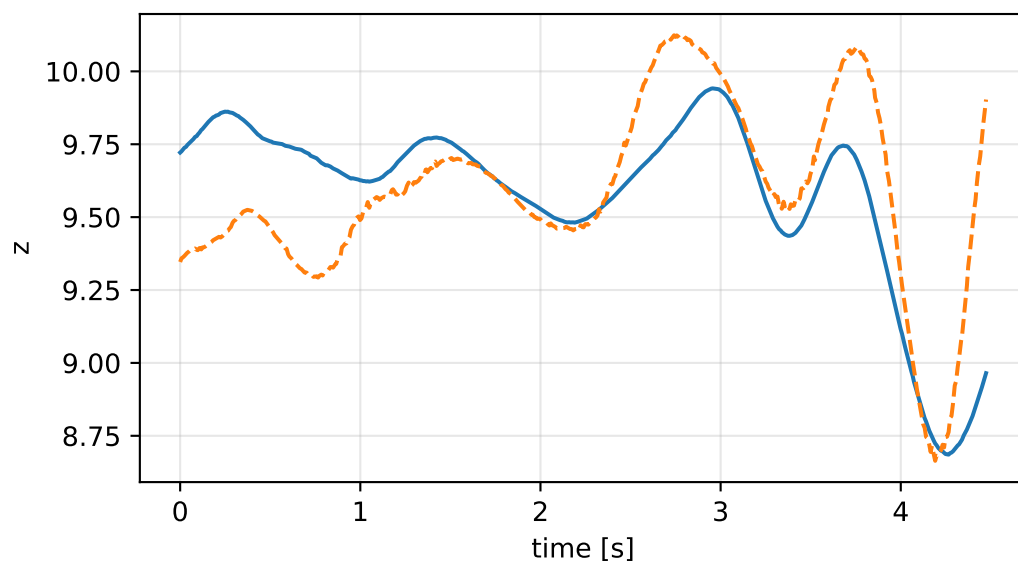
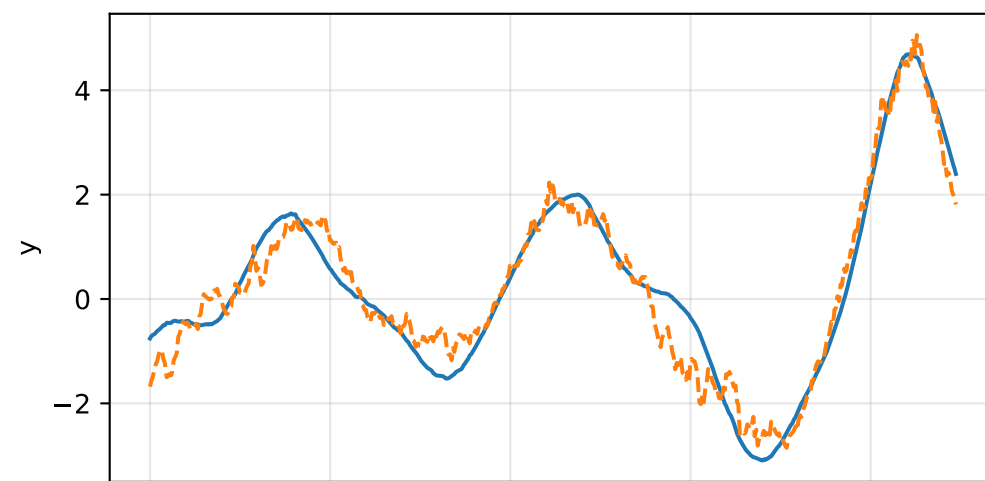
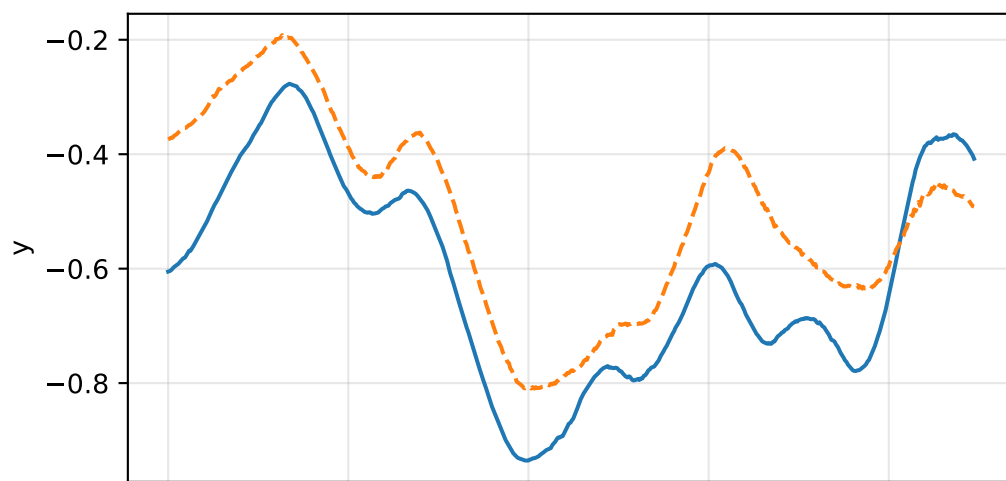
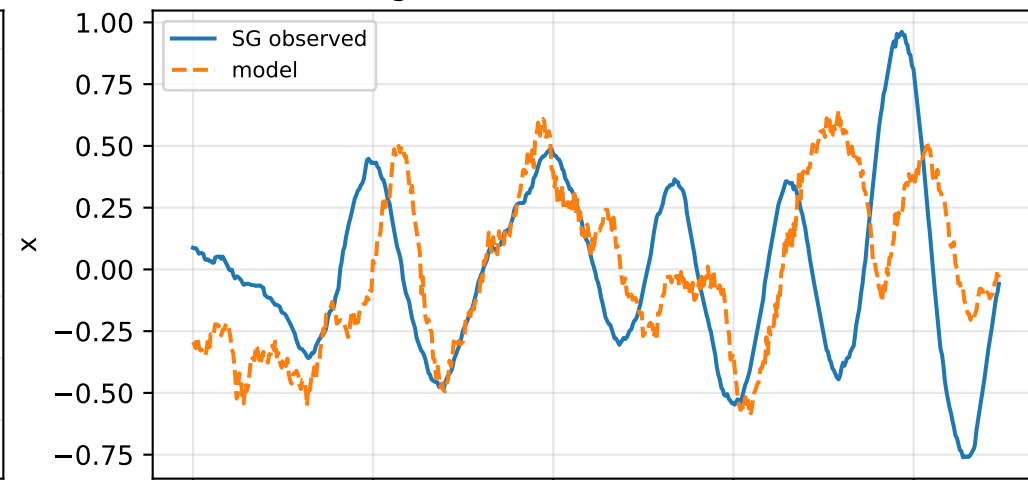


inertia_over_mass_all_nominal: acceleration residual objective

specific acceleration [m/s^2]

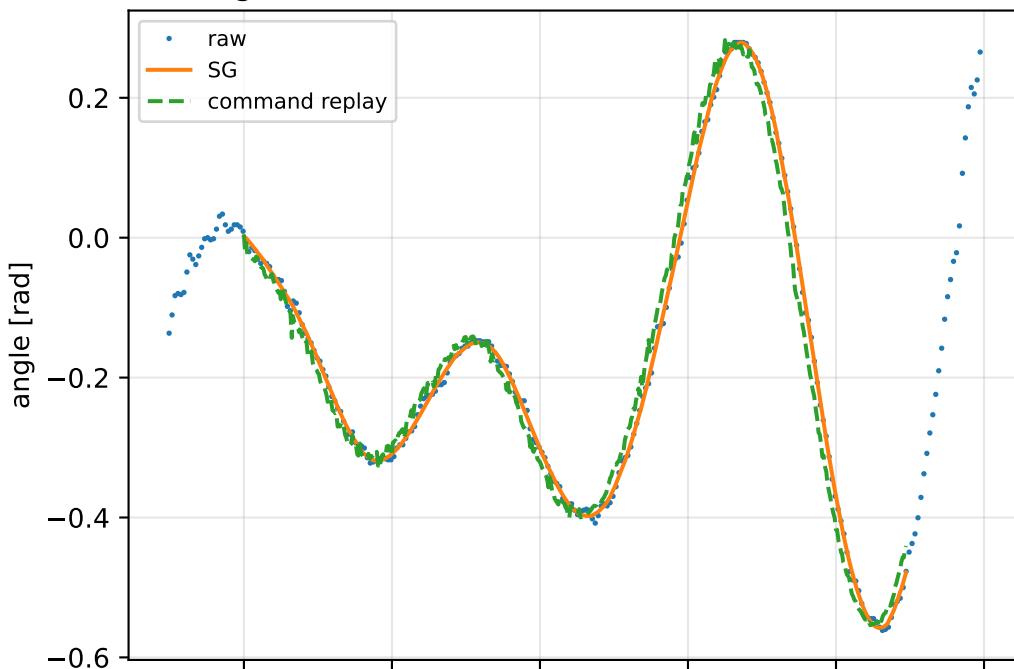


angular acceleration [rad/s^2]

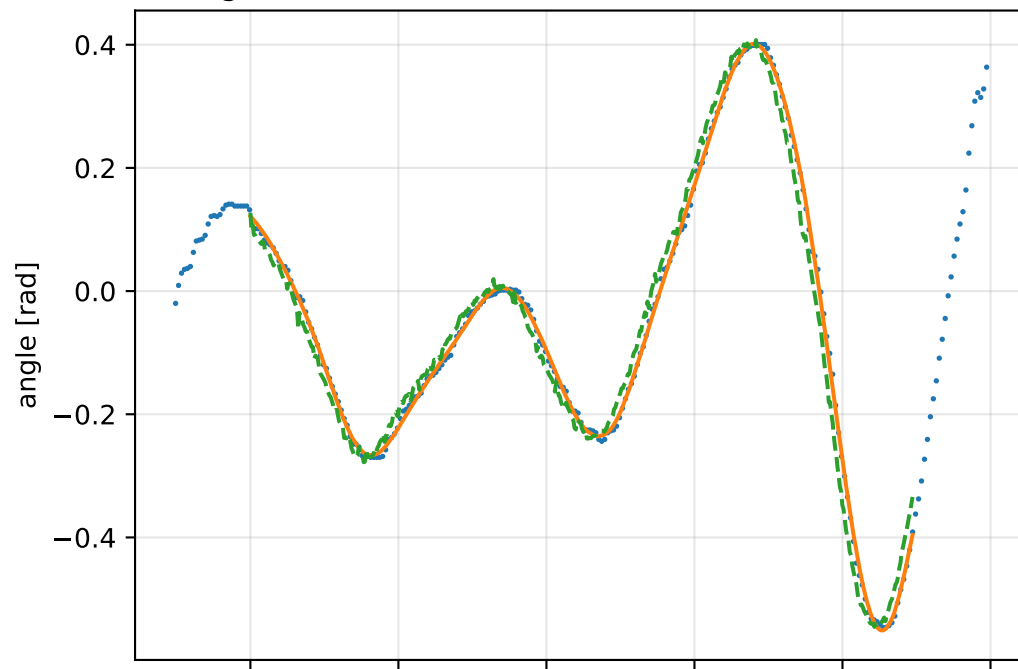


inertia_over_mass_all_nominal: gimbal measurement audit (objective=measured_sg)

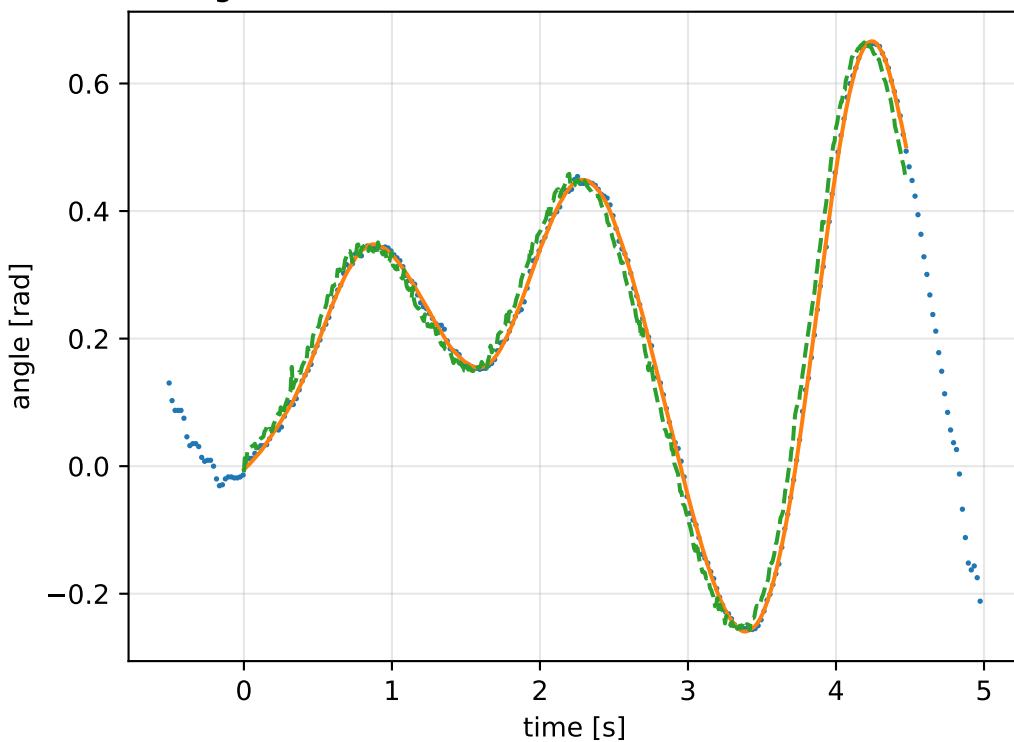
gimbal 1: RMSE=0.027, max=0.07529 rad



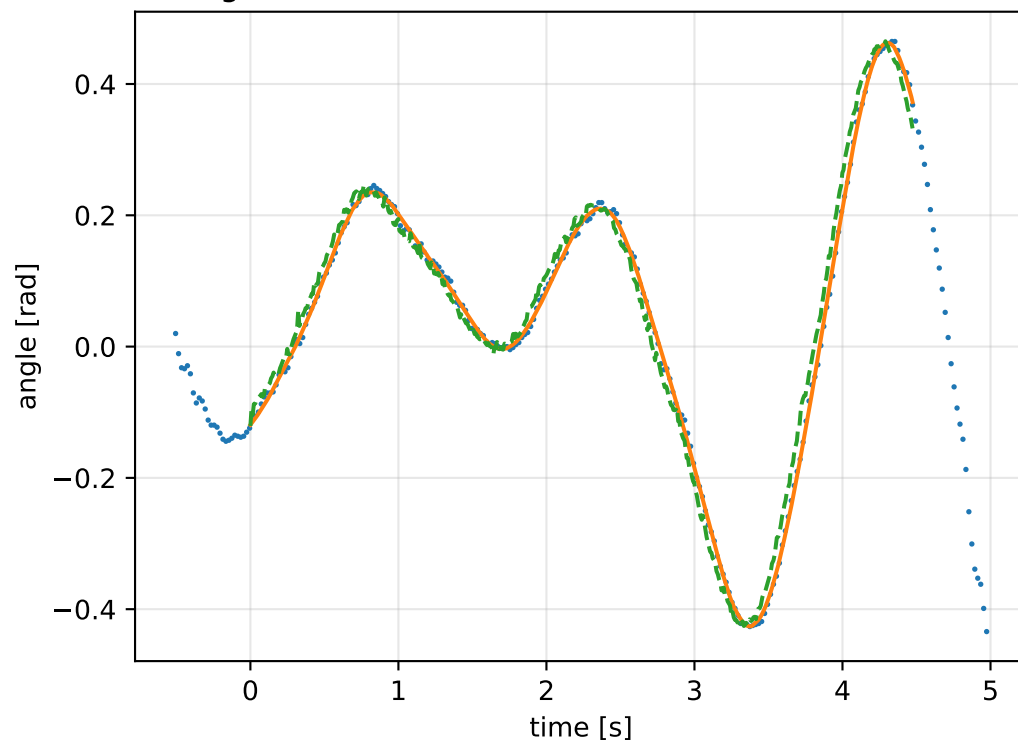
gimbal 2: RMSE=0.03107, max=0.08239 rad



gimbal 3: RMSE=0.03106, max=0.08682 rad

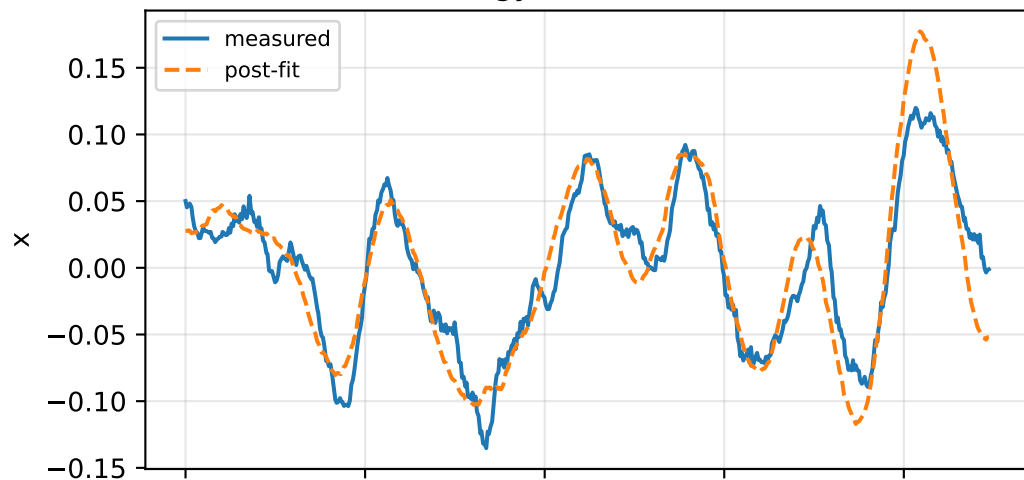


gimbal 4: RMSE=0.02598, max=0.07261 rad

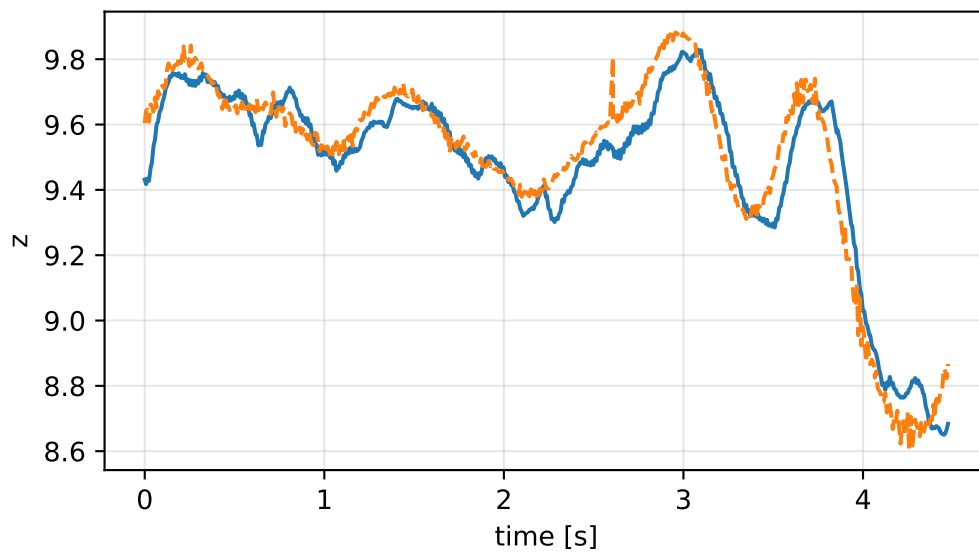
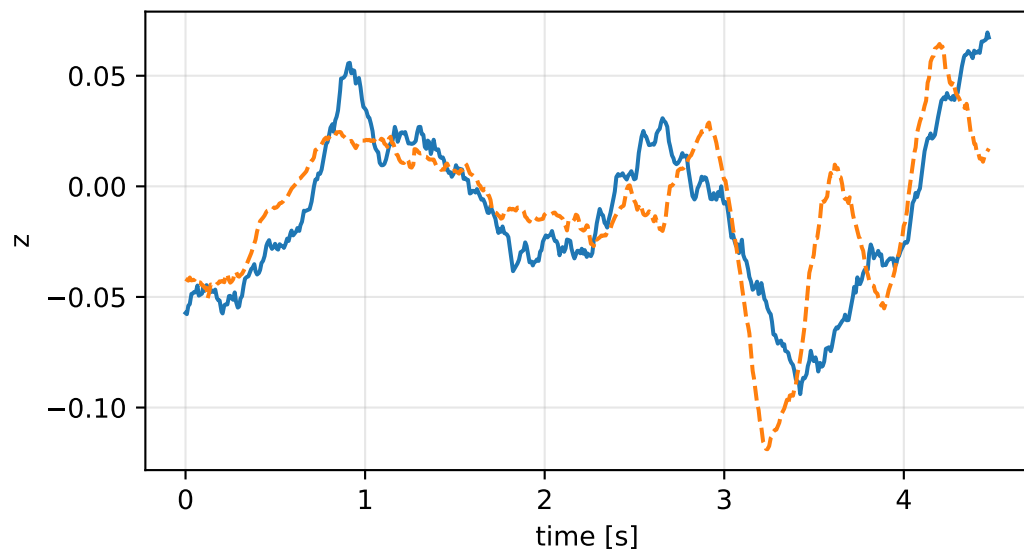
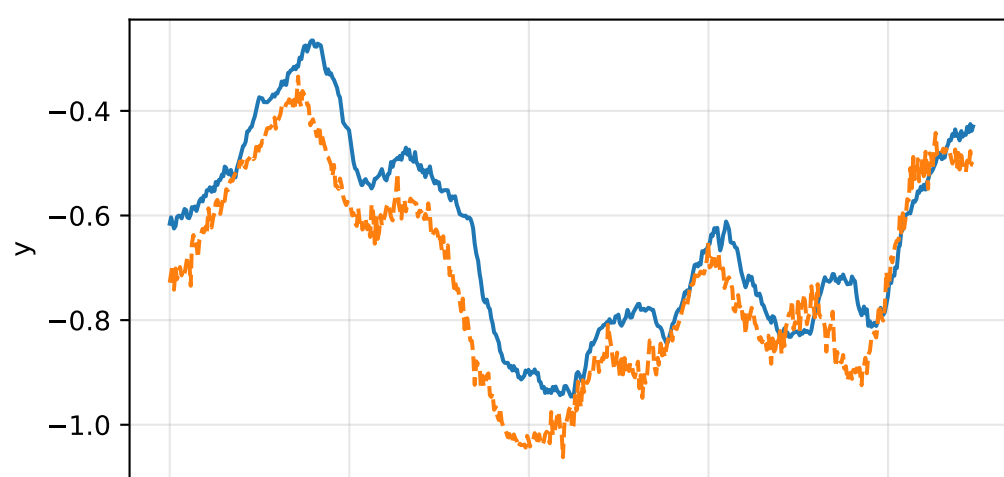
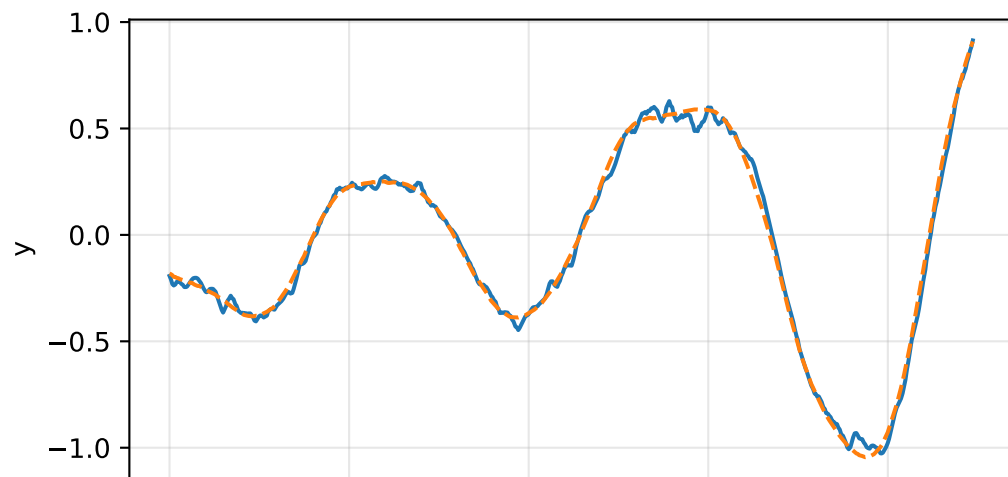
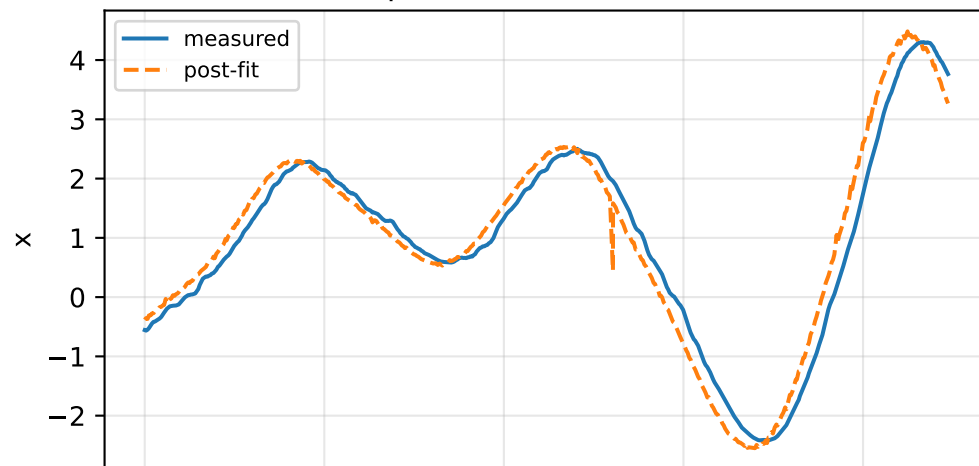


inertia_over_mass_all_nominal: IMU comparison (diagnostic only)

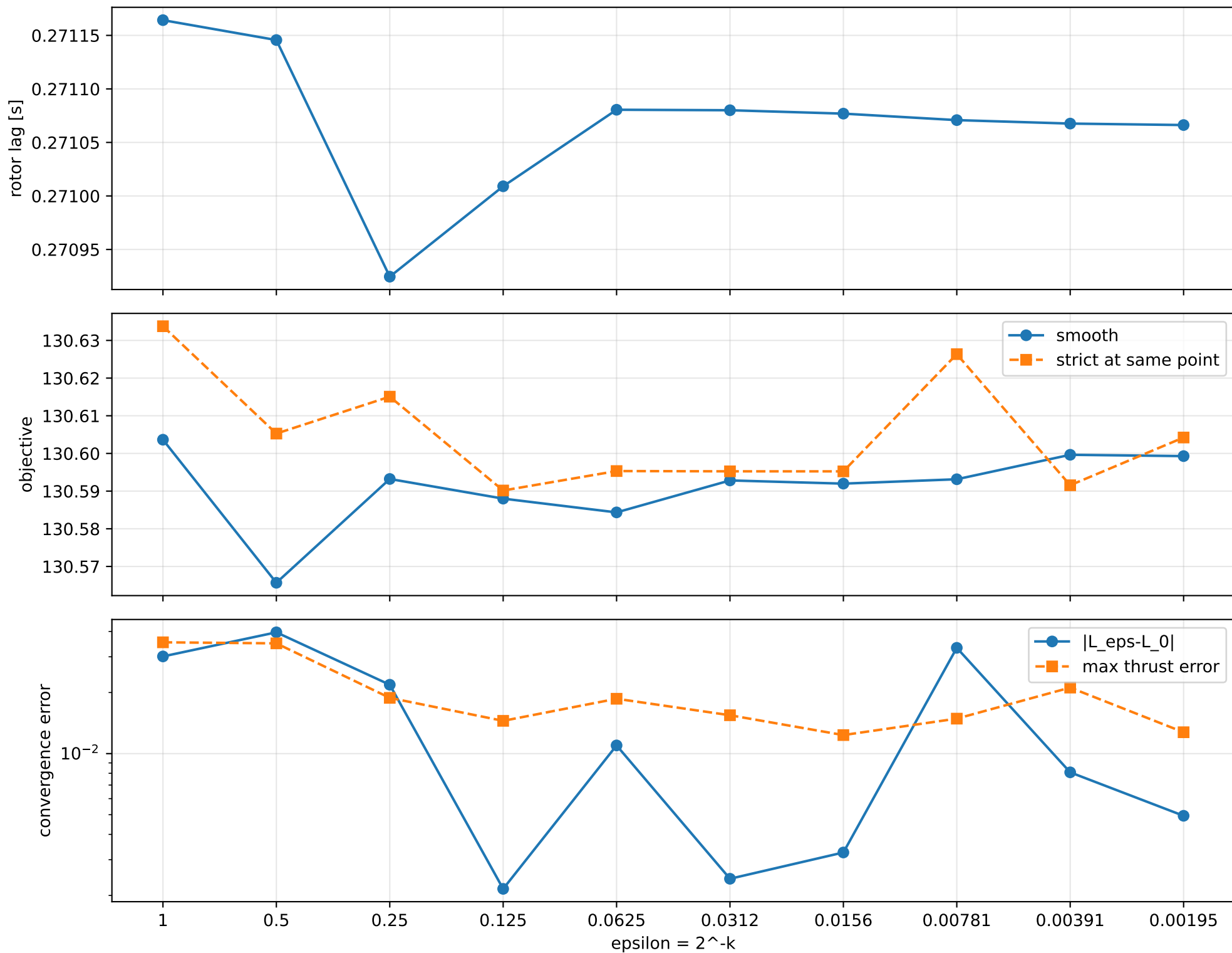
gyro [rad/s]



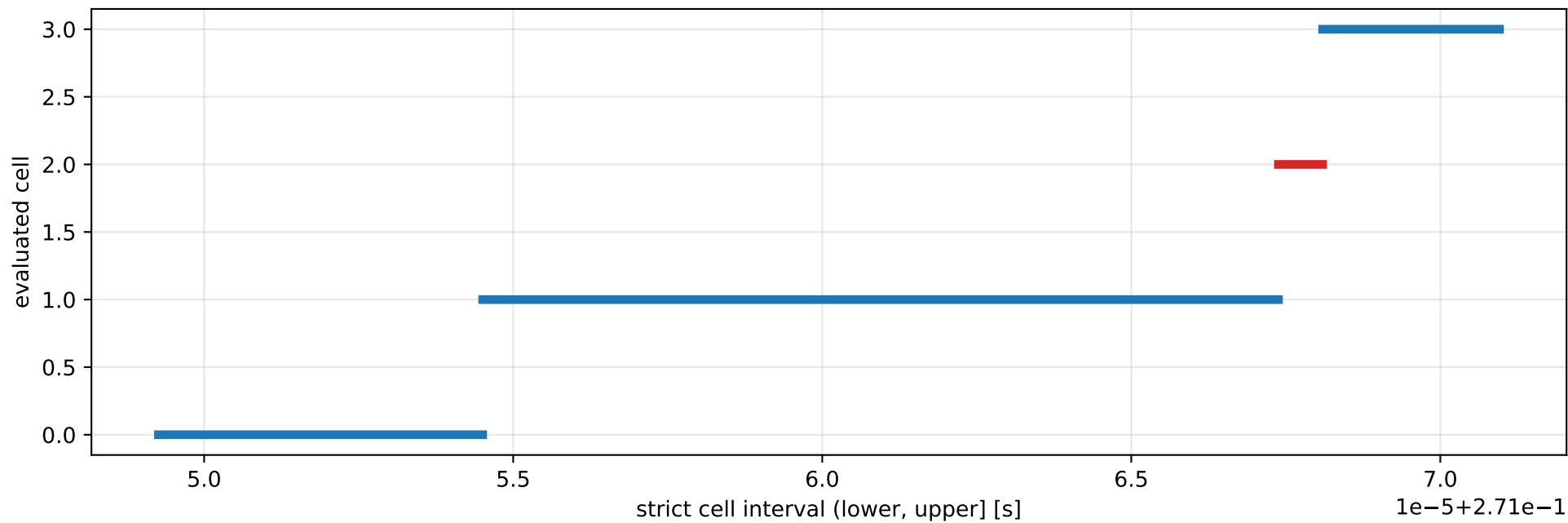
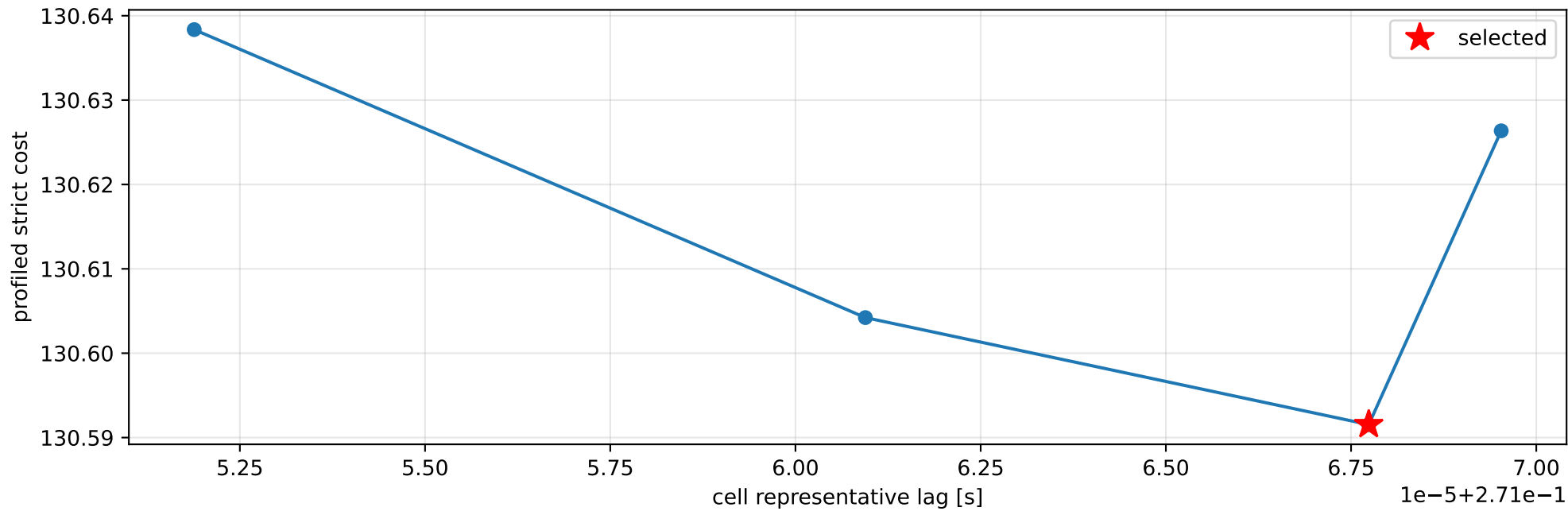
specific force [m/s^2]



inertia_over_mass_all_nominal: smooth-to-strict continuation



inertia_over_mass_all_nominal: exact strict-ZOH lag cells



selected (0.271067381, 0.271068096] s; final smooth 0.271066244 s; data boundary=False

inertia_over_mass_all_nominal: parameter estimate

Case: inertia_over_mass_all_nominal

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 2.55463729

inertia [kg m²]:

[[7.06122463e-02 -7.86784861e-07 2.06558865e-05]

[-7.86784861e-07 7.05607474e-02 6.25832885e-08]

[2.06558865e-05 6.25832885e-08 1.40129442e-01]]

CoG body [m]: [-0.01215173 0.00068531 0.01707533]

force effectiveness: [0.96205312 0.88714978 0.88827235 0.9470891]

scale-free J/m [m²]:

[[2.76408109e-02 -3.07983001e-07 8.08564352e-06]

[-3.07983001e-07 2.76206519e-02 2.44979155e-08]

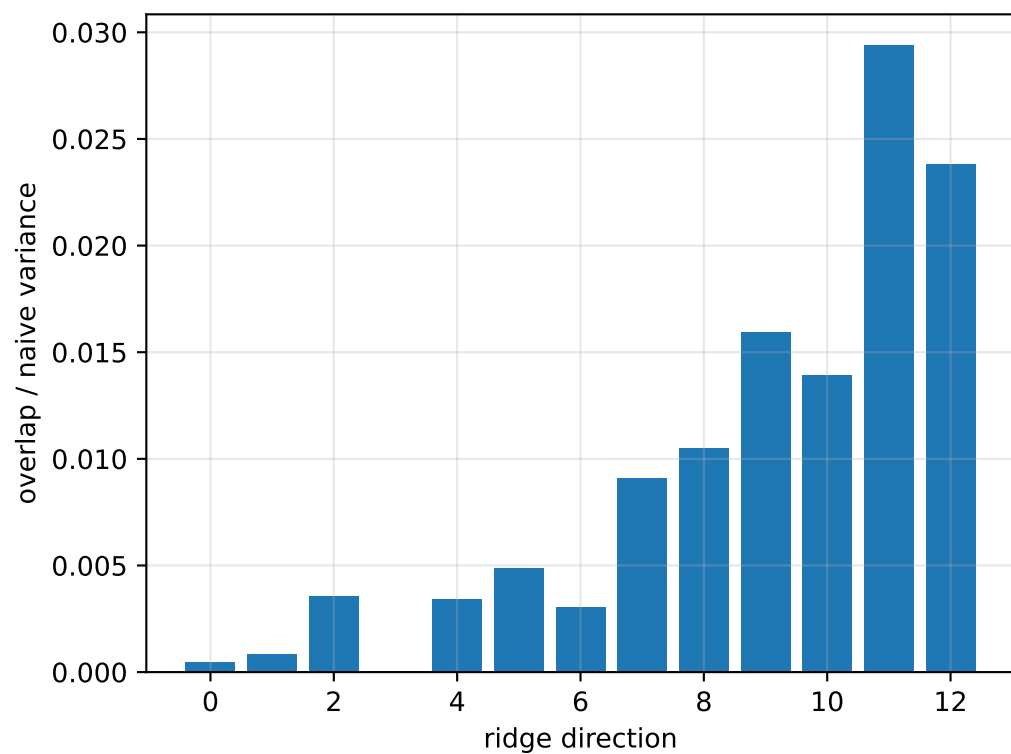
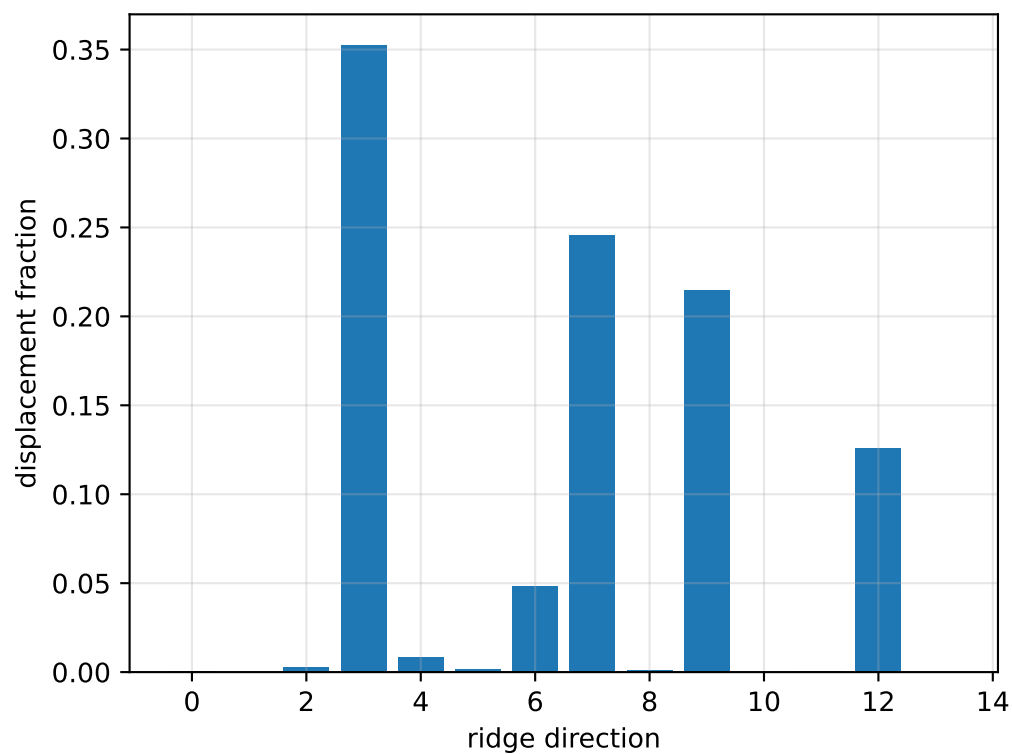
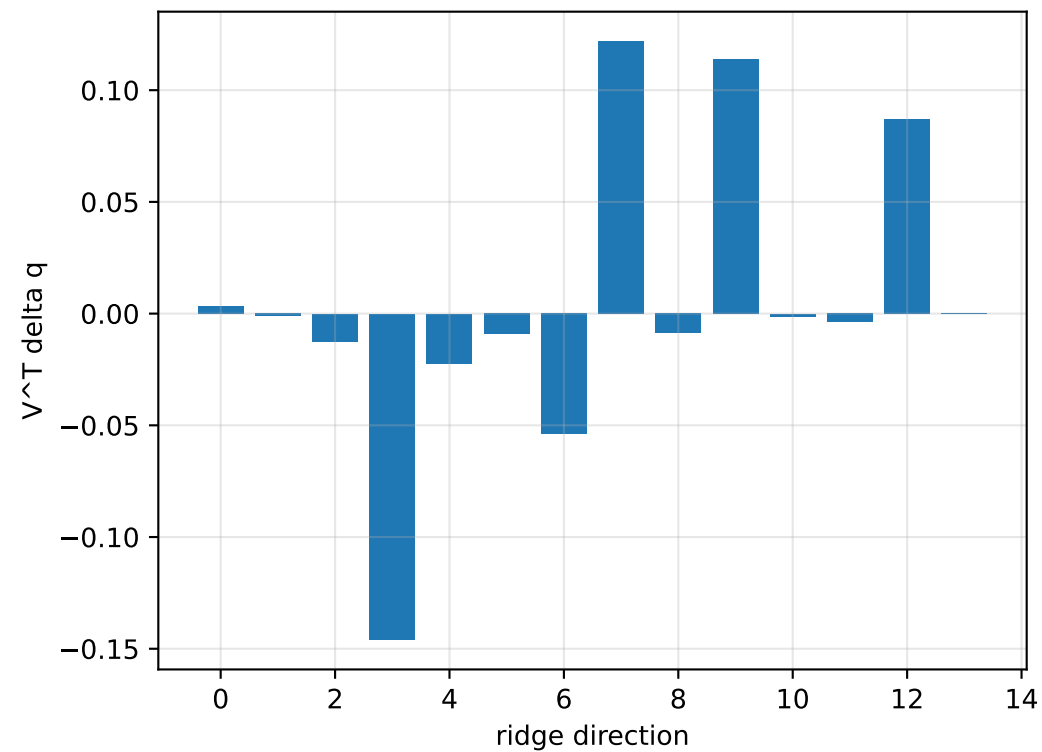
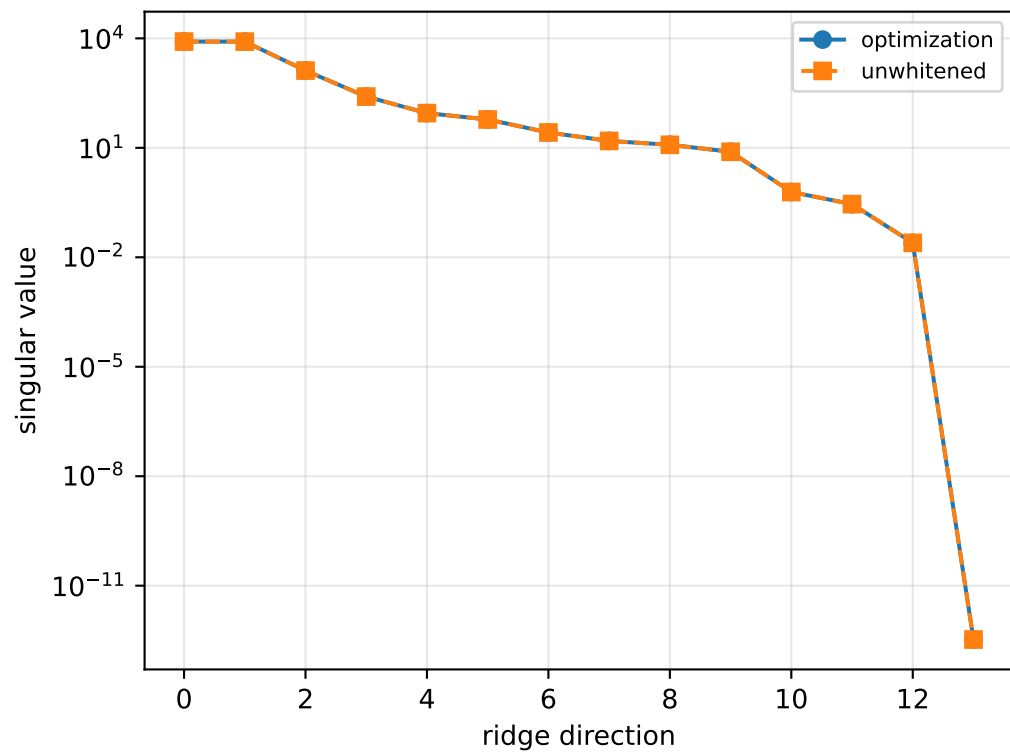
[8.08564352e-06 2.44979155e-08 5.48529695e-02]]

scale-free f/m: [0.37659089 0.34727035 0.34770977 0.3707333]

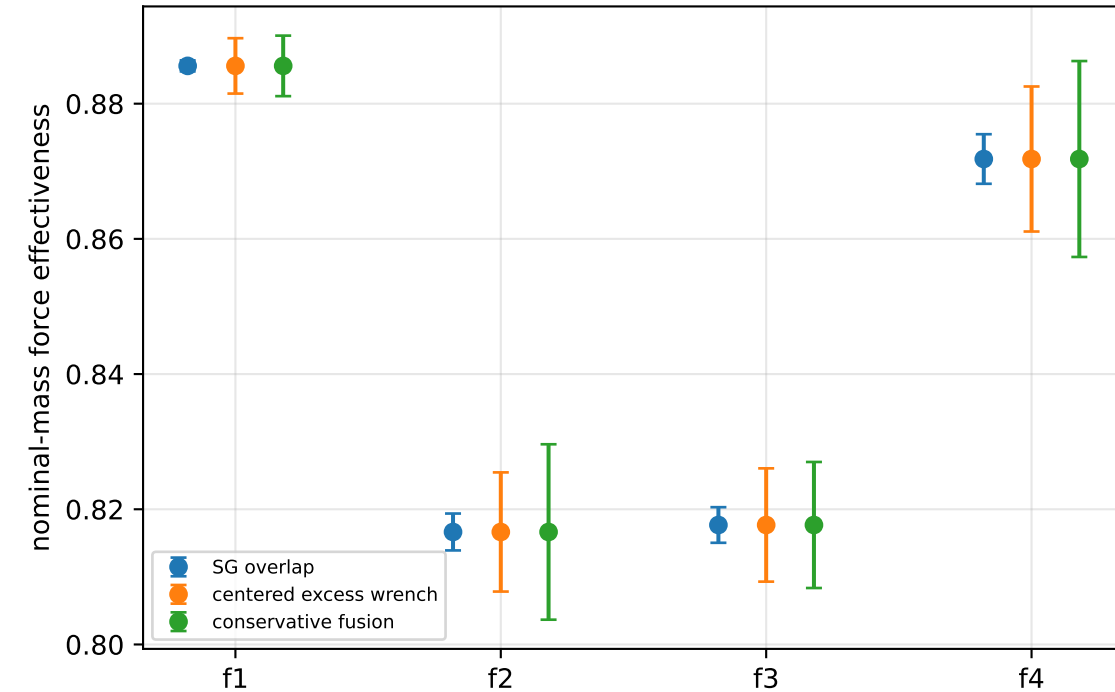
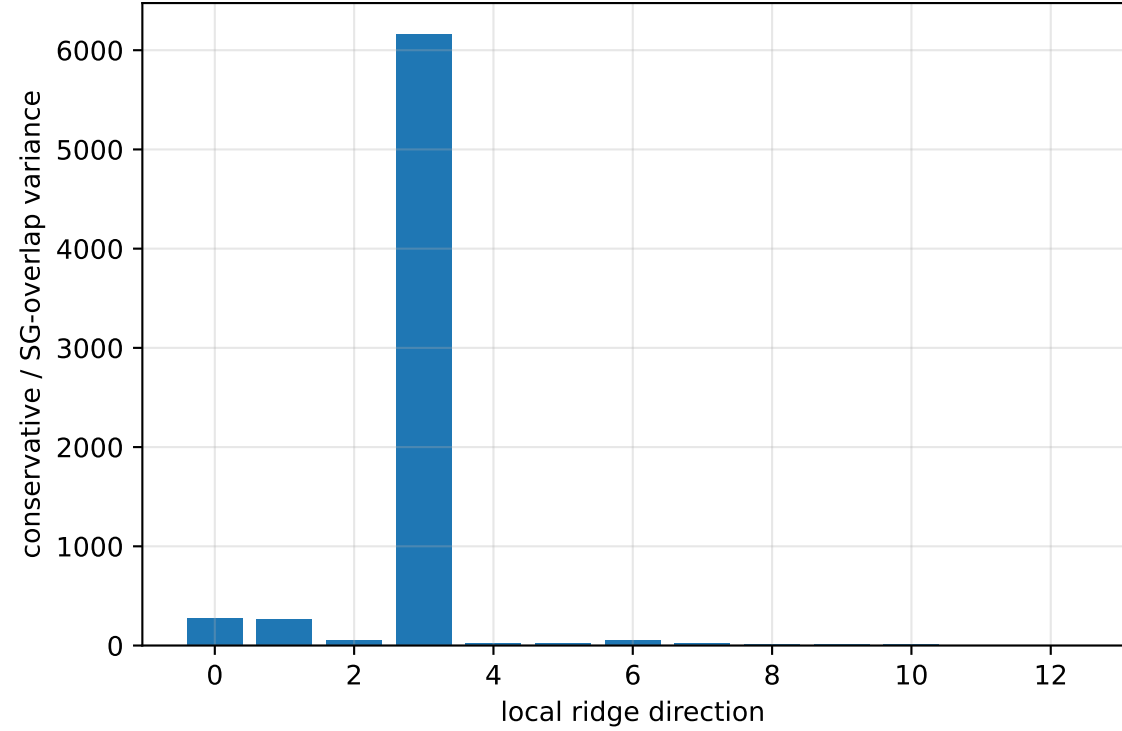
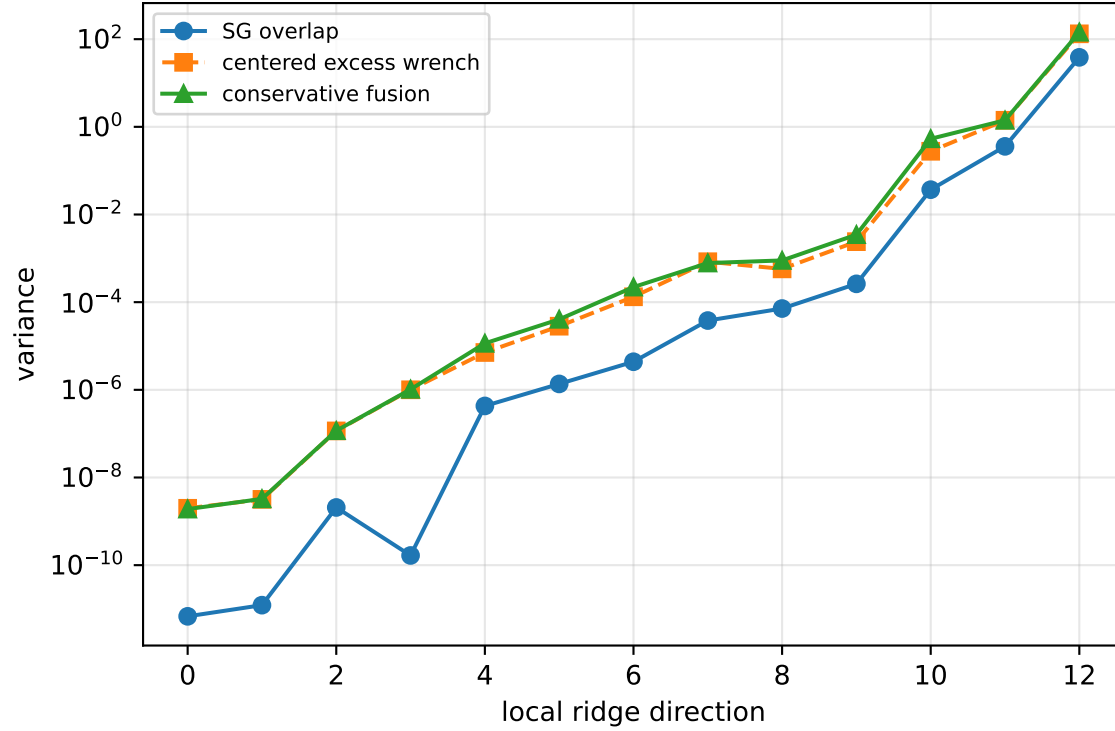
rotor lag cell: (0.271067381, 0.271068096] s

representative: 0.271067739 s

inertia_over_mass_all_nominal: ridge spectrum (rank 13, nullity 1, ||jv||=2.36e-13)



inertia_over_mass_all_nominal: Conservative fusion uncertainty



Top three non-gauge ambiguous directions
 $\text{tr}(\mathbf{M}_{\text{res}}) / \text{tr}(\mathbf{M}_{\text{SG}}) = 255.1$
wrench-to-acceleration recovery max error = $2.89\text{e-}15$
exact scale gauge ridge index = 13 (alignment 1)

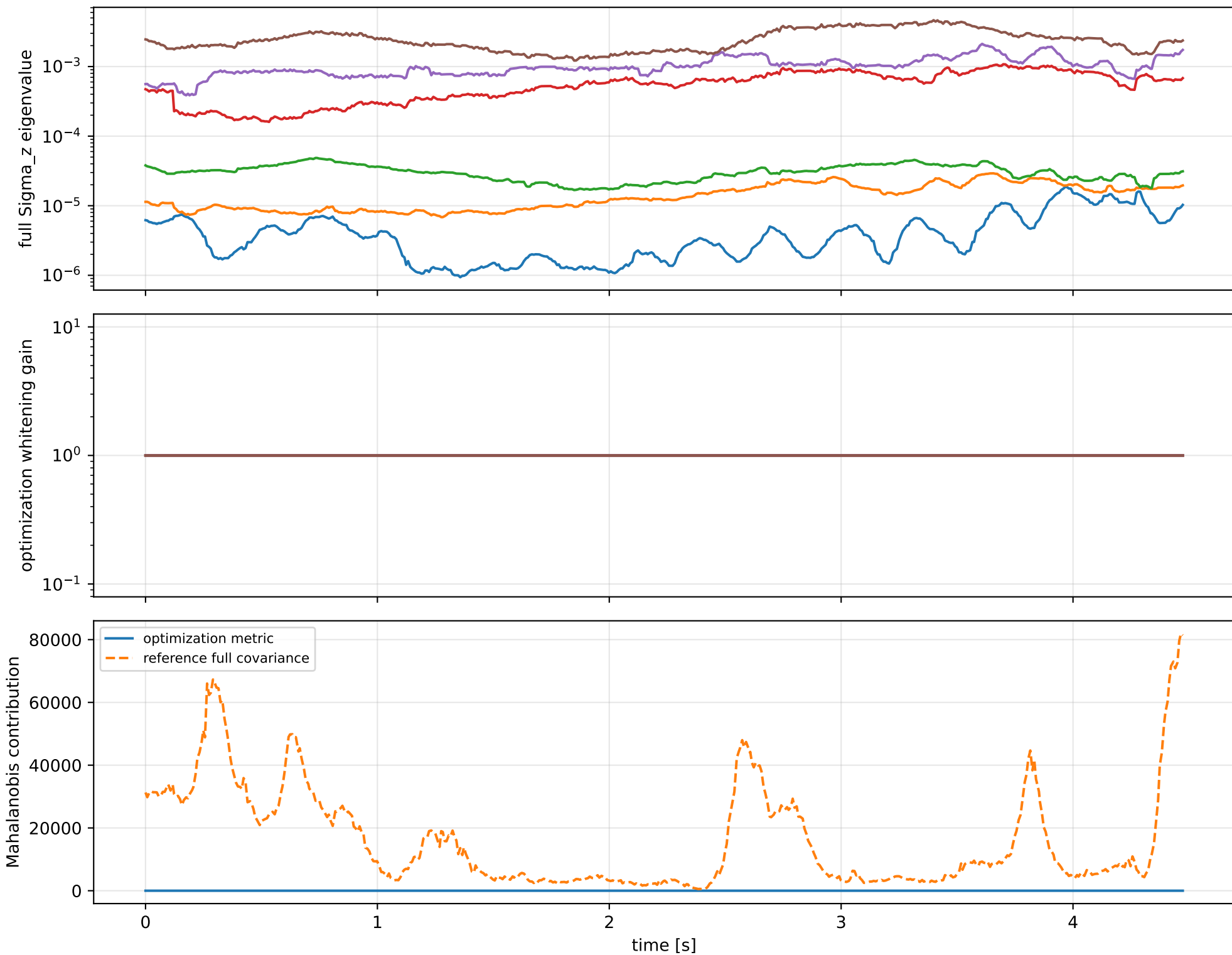
index 12: variance=143.7, ratio=3.732
 $\log_{\text{mass}} = -0.131$; second_moment_diag_1=-0.139
second_moment_diag_2=-0.139; second_moment_diag_3=+0.935
second_moment_offdiag_12=-0.000305; second_moment_offdiag_13=+0.0253
second_moment_offdiag_23=+0.00674; cog_x=-2.47e-05
cog_y=+2.11e-05; cog_z=-6.25e-06
log_force_eff_1=-0.132; log_force_eff_2=-0.132
log_force_eff_3=-0.132; log_force_eff_4=-0.131

index 11: variance=1.413, ratio=3.943
 $\log_{\text{mass}} = +0.00555$; second_moment_diag_1=+0.00484
second_moment_diag_2=-0.00836; second_moment_diag_3=-0.0241
second_moment_offdiag_12=+0.000568; second_moment_offdiag_13=+0.998
second_moment_offdiag_23=+0.0558; cog_x=-0.00011
cog_y=+0.000115; cog_z=+3.81e-05
log_force_eff_1=+0.00567; log_force_eff_2=+0.00437
log_force_eff_3=+0.00564; log_force_eff_4=+0.00637

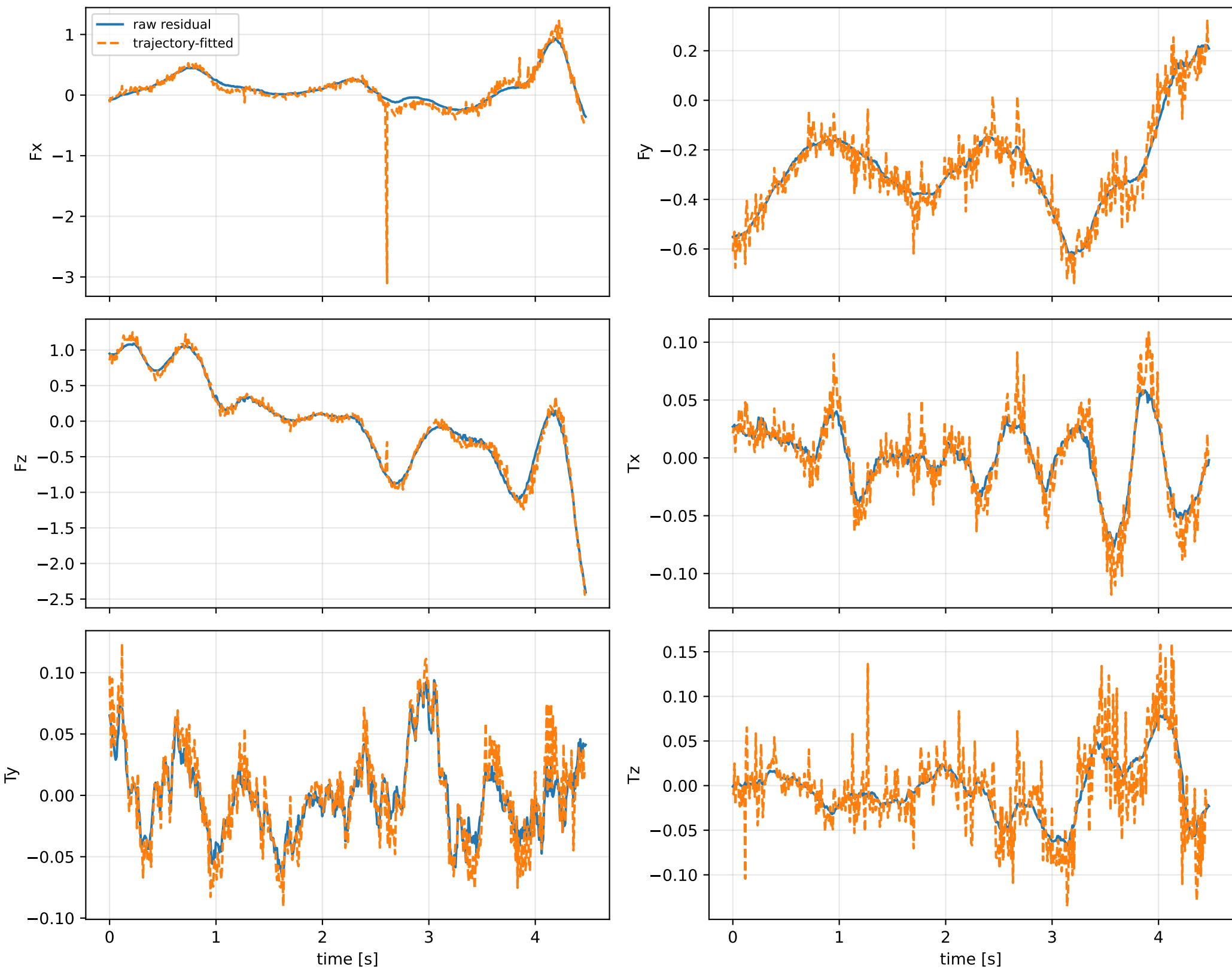
index 10: variance=0.5333, ratio=14.47
 $\log_{\text{mass}} = +0.00108$; second_moment_diag_1=+0.00295
second_moment_diag_2=-0.0143; second_moment_diag_3=+0.00491
second_moment_offdiag_12=-0.00843; second_moment_offdiag_13=+0.0558
second_moment_offdiag_23=-0.998; cog_x=+0.000665
cog_y=-0.00163; cog_z=-0.000108
log_force_eff_1=+0.00244; log_force_eff_2=+0.015
log_force_eff_3=-0.00619; log_force_eff_4=-0.00587

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.

inertia_over_mass_all_nominal: optimization vs reference covariance



inertia_over_mass_all_nominal: wrench / closure (force max 7.22e-15, torque max 1.06e-16)



inertia_over_mass_all_nominal: optional parameter-prior audit

Optional physical parameter prior

The parameter prior is optional external information. The default estimator is prior-free.

name: pseudo_condition_inertia_over_mass_all_nominal

role: pseudo_conditioning_ablation

source: /home/leus/catkin_ws/src/ProbTF-demo/ros/examples/grape-param-estim/minimal/config/priors/pseudo_conditioning/group/inertia_over_mas

SHA256: 696c77257c9aa3a12bfed1ee5eb82a98ad78c606342ec66cb8ccb4cf17bb8a36

data objective: 130.59153796

prior objective: 2.61347845122e-06

total objective: 130.591540573

This configuration uses an intentionally tight artificial standard deviation for an ablation-style pseudo-conditioning experiment; it is not a calibrated physical uncertainty.

factor: inertia_over_mass_all_nominal (inertia_over_mass_m2)

components: xx, yy, zz, xy, xz, yz

target: [2.76408097e-02 2.76206510e-02 5.48529694e-02 -3.09533934e-07
8.08602633e-06 2.51604719e-08]

std: [1.e-06 1.e-06 1.e-06 1.e-06 1.e-06 1.e-06]

final: [2.76408109e-02 2.76206519e-02 5.48529695e-02 -3.07983001e-07
8.08564352e-06 2.44979155e-08]

error: [1.17937219e-09 9.05502163e-10 1.58716533e-10 1.55093272e-09
-3.82805188e-10 -6.62556364e-10]

standardized residual: [0.00117937 0.0009055 0.00015872 0.00155093 -0.00038281 -0.00066256]

factor objective: 2.61347845122e-06